

Free Paper Session

PATIENT PERSPECTIVES OF ACCEPTABILITY OF RISK-BASED INDIVIDUALIZED DIABETIC RETINOPATHY

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Design: In-depth ethnographic qualitative study.

Purpose: There is increasing worldwide interest in developing variable interval protocols in screening for sight threatening diabetic retinopathy (DR). However the acceptability and therefore safety of a change from annual intervals has not been well evaluated, particularly from a patient perspective. We used in-depth qualitative methods to develop a detailed, contextualized understanding of patient perspectives of introducing individualized risk-based screening intervals.

Methods: People with diabetes (PWD) who attend for DR screening were recruited through primary care providers, using purposive sampling. Qualitative in-depth semi-structured interviews following a grounded theory method explored patient experiences of screening, diabetes care, and views on individualized variable screening intervals.

Results: In 34 interviews, PWD expressed a range of views. Many agreed in principle to intervals being individualized and risk-based. However, some considered a 24 month interval to be too long. Some felt that extended durations would introduce an unacceptable level of risk of developing DR. Some felt annual eye screening offers an important role in reassuring patients of good eye health. Some found the screening appointments unpleasant and inconvenient but would rather attend more frequently so that DR could be detected.

In contrast, many recognized the need for NHS resources to be concentrated where they are most needed but suggested that if extended intervals are introduced, safeguards must be put in place to enable patients to re-enter the screening system should risk factors worsen.

Some highlighted that being assigned to particular risk groups may impact upon their likelihood to attend screening appointments and perceptions of the importance of general diabetes management.

Poor levels of understanding of DR, eye screening and risk appeared to influence patient perspectives.

Conclusions: Findings suggest that for individualized screening programs to be successful there is a need for improved patient education surrounding diabetes related eye health and specifically around the risks and benefits associated with extended intervals.

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DIABETIC RETINOPATHY IN EUROPE INITIAL RESULTS OF THE E3 CONSORTIUM DIABETES GROUP

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Design: Pan-European cross-sectional study as part of the European Eye Epidemiology (E3) Consortium Diabetes Study Group program.

Purpose: The purpose of this study is to estimate the prevalence and review risk factors for diabetic retinopathy (DR) in Europe, and to investigate the presence or absence of regional variations between European countries.

Methods: E3 consists of 24 study groups from all regions of Europe. E3 members were invited to participate in the DR study by sharing data and images from patients involved in their studies. Cohort studies as well as case-control and case-only studies are all included. For cohorts, where patients are not graded for DR the images are being graded either by the study group or by the Reading Centre at Moorfields Eye Hospital. Grading is a combination of human and semi-automated grading utilizing the University of Surrey software.

Results: Only 2 studies have no relevant data at present, the rest of the E3 are willing to participate. Of those with relevant data, results have been received from 10 studies involving 102,997 patients. The rest of the studies are in image analysis or data analysis phase within the cohort. Analysis of the 6 case-only studies shows a total population of 68,483 patients of which 50.75% was men. In the 6 case-only studies 4.95% had Type 1 diabetes, 22.61% had Type 2 and 72.43% was either unknown or not collected in the studies population. Duration of diabetes between 1-20 years accounts for 75% of the sample size. The first analysis shows that 73% of subjects have no DR, 22.9% have non-proliferative DR and 1.5% have proliferative DR, 2.6% was either not gradable or image could not be taken.

Conclusions: This is the first pan-European DR study of such scale. The first results show agreement with prior knowledge of most patients having Type 2 diabetes mellitus and having no DR. The results will be useful for establishing and planning further development of diabetic eye services in the relevant regions.

INCIDENCE AND PROGRESSION OF RETINOPATHY AND VISUAL IMPAIRMENT IN PEOPLE WITH DIABETES IN SUB-SAHARAN AFRICA; RELATIONSHIP WITH POPULATION SPECIFIC VARIABLES

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Design: Prospective cohort study.

Purpose: Sub-Saharan Africa faces an epidemic of diabetes. We performed the first prospective cohort study of diabetic retinopathy (DR) in this region. We report progression of DR and associations with demographic, clinical and biochemical variables in people with diabetes in Southern Malawi.

Methods: Subjects were systematically sampled from two hospital-based diabetes clinics providing primary diabetes care to a population affected by high levels of infectious disease. Clinical examination and biochemical testing was performed to assess visual acuity, glycemic control, systolic BP, HIV status, urine albumin-creatinine ratio, hemoglobin and serum lipid levels. DR was graded using modified Wisconsin grading at an accredited reading center. Sight-threatening diabetic retinopathy (STDR) was defined as moderate pre-proliferative DR or worse, circinate maculopathy or exudates within one disc diameter of the foveal center.

Results: Of 357 subjects recruited 295 were assessed at 24 months, 28 were confirmed dead (90.5% follow-up). At baseline 13.4% subjects were HIV-positive and 15.1% anemic. At follow-up (median 2.0 yrs) rates of progression were: 2 step (or greater) 58/293 (19.8%); STDR 23/225 (10.2%). Cumulative incidence at 24 months of STDR for subjects with Level 10 (no retinopathy), Level 20 (background) and Level 30 DR at baseline were 2.7% (95% CI 0.1-5.3), 27.3% (16.4-38.2) and 25.0% (0-67.4), respectively. In multivariate logistic analysis 2 step progression of DR was associated with HbA1c (OR 1.27, 95% CI 1.12-1.45), baseline grade of DR (1.39, 1.02-1.91) and HIV infection (OR 0.16, 0.03-0.78). 85 persons required laser photocoagulation. Rates of progression to visual loss were: >15 letters lost 17 subjects (5.8%), moderate visual impairment (<60 letters) 3 (1.0%), severe visual impairment (<50 letters) 5 (1.7%).

Conclusions: Progression to STDR from no DR and background DR occurred approximately 3 and 2.5 times more frequently than reported in recent European studies, respectively. The negative association of HIV infection with DR progression is a new finding. Our results highlight the urgent need for provision of services for DR detection and management to avoid a large burden of vision loss.

REGISTRY STUDY OF DIABETIC MACULAR EDEMA IN TURKEY: TURK-DEM STUDY

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Design: National, multicenter, observational study.

Purpose: To determine characteristics of newly diagnosed patients with diabetic macular edema (DME) and thereby to identify the disease characteristics and local treatment approaches.

Methods: This study was conducted at 38 centres representing all geographic areas in Turkey. Patients aged $\geq$ 18 years and newly diagnosed with DME but had not received any treatment specific for DME were enrolled. Patients' data regarding demographic characteristics, vital signs, body mass index, general and ocular medical history, comorbid diseases, history for and clinical findings related to diabetes, routine biochemical test results, DME type and central macular thickness (CMT), as well as diagnostic methods and treatment plans of the centres were recorded on electronic case report forms.

Results: This study included 1890 eyes of 945 patients (mean age, 61.3 ± 9.9 and male, 55.2%), of whom 96.4% had type II diabetes. Time after diagnosis was >5 years in 88% of the patients. With regard to diabetes treatment, 48.3% were on insulin and 43.9% on oral antidiabetics; 20.7% received no treatment. At least one comorbid disease was present in 80%. The reason for admission to ophthalmology departments was visual problems in 52.2%. DME diagnosis was confirmed by fundus fluorescein angiography in 73.0% and optical coherence tomography in 98.8% of the patients. Diffuse, focal, and mixed DME was present in 39.2%, 36.9%, and 9.7% of the eyes, respectively. The rate of CMT >300 μm was 66.6%. Primary treatment plans were laser photocoagulation and/or anti-vascular endothelial growth factor therapy. The rate of receiving